

```

print("Program 1")
print("Hello World")
print ("")

print("Program 2")
usertext = input("What is your name? ")
print("Hello", usertext)
print ("")

print("Program 3")
num1 = input('Enter first number: ')
num2 = input('Enter second number: ')
sum = float(num1) + float(num2)
print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))
print ("")

print("Program 4")
num1 = input('Enter first number: ')
num2 = input('Enter second number: ')
average =(int(num1) + int(num2))
print('average:{0} '.format(average))
print ("")

print("Program 5")
visagrade = input('enter your visa grade : ')
finalgrade = input('enter your final grade : ')
average =(float(visagrade)*0.3)+(float(finalgrade)*0.7)
print("average :{0} ".format(average))
print ("")

print("Program 6")
firstexam = input('your first exam : ')
secondexam = input('your second exam : ')
thirdexam = input('your third exam : ')
average =(float(firstexam)+float(secondexam)+float(thirdexam))/3
print("average :{0} ".format(average))
print ("")

print("Program 7")
average = input('enter average : ')
if(int(average)>=50):
    print("Passed")
else:
    print("Failed")
print ("")

```

```
print("Program 8")
num = int(input("Enter a number: "))
if (num % 2) == 0:
    print("{0} is Even".format(num))
else:
    print("{0} is Odd".format(num))
print ("")
```

```
print("Program 9")
num = float(input("Enter a number: "))
if num > 0:
    print("Positive number")
elif num == 0:
    print("Zero")
else:
    print("Negative number")
print ("")
```

```
print("Program 10")
print("body mass index calculation program")
height = float(input("enter height (m):"))
weight = int(input("enter weight (kg):"))
index = weight/(height*height)
if index <=18:
    print("\n underweight BMI:{0}".format(index))
elif index > 18 and index <=25 :
    print("\n overweight BMI:{0}".format(index))
elif index > 25 and index <=30:
    print("\n obese BMI:{0}".format(index))
elif index > 30:
    print("\n severely obese BMI:{0}".format(index))
print ("")
```

```
print("Program 11")
age = input('enter age : ')
if(int(age)<18):
    print("Your Age Is Not Eligible To Get A Driver's License")
else:
    print("Your Age Is Eligible To Get Your License")
print ("")
```

```
print("Program 12")
for i in range(1,101):
    print(i)
print ("")
```

```
print("Program 13")
for i in range(1,101):
    if i%2==0:
        print(i)
print ("")
```

```
print("Program 14")
for i in range(1,101):
    if i%2!=0:
        print(i)
print ("")
```

```
print("Program 15")
for i in range(1,101):
    if i%3==0 or i%5==0:
        print(i)
print ("")
```

```
print("Program 16")
num = input('enter number : ')
for i in range(1,int(num)+1):
    print(i)
print ("")
```

```
print("Program 17")
short = input('Enter short side : ')
tall = input('Enter tall side : ')
area = int(short) * int(tall)
perimeter = 2 * (int(short) + int(tall))
print("area: {0}".format(area))
print("perimeter: {0}".format(perimeter))
```

```
print ("")
```

```
print("Program 18")
word = 'mrhuseyin'
for char in word:
    print(char)
print ("")
```

```

print("Program 19")
sumofnumbers = 0
num1 = input('first number: ')
num2 = input('second number: ')
for i in range(int(num1) + 1, int(num2)):
    sumofnumbers += i

print("Sum of numbers between {0} and {1} : {2}".format(num1, num2,
sumofnumbers))

```

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print ("")

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```

print("Program 20")
selection = input("Press (1) for Cinema, (2) for Theater : ")
student = input("Are you student(Y/N) : ")
price = 0
#non-discounted fee calculation
if selection == '1':
    price = 10 #cinema
elif selection == '2':
    price = 5 #theatre
#student discount
if student == 'Y' or student == 'y':
    price=price / 2 #%50
    print(" The fee you have to pay :{}".format(price))
print ("")

```

```

print("Program 21")
num = int(input("Enter a number: "))

```

```

if num > 1:
    for i in range(2,num):
        if (num % i) == 0:
            print(num,"is not a prime number")
            print(i,"times",num//i,"is",num)
            break
    else:
        print(num,"is a prime number")

```

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else:
    print(num,"is not a prime number")
print ("")

```

```

print("Program 22")
NumList = []

```

```

Even_Sum = 0
Odd_Sum = 0

Number = int(input("Please enter the Total Number of List Elements: "))
for i in range(1, Number + 1):
    value = int(input("Please enter the Value of %d Element : " %i))
    NumList.append(value)

for j in range(Number):
    if(NumList[j] % 2 == 0):
        Even_Sum = Even_Sum + NumList[j]
    else:
        Odd_Sum = Odd_Sum + NumList[j]

print("\nThe Sum of Even Numbers in this List = ", Even_Sum)
print("The Sum of Odd Numbers in this List = ", Odd_Sum)
print ("")

print("Program 23")
newsalary = 0
salary = input("enter new salary : ")
raise_rate = input("salary raise rate(%) : ")
newsalary = int(salary) + (int(salary) * int(raise_rate) / 100)
print("increased salary :",newsalary)
print ("")

print("Program 24")
import math

def find_Diameter(radius):
    return 2 * radius

def find_Circumference(radius):
    return 2 * math.pi * radius

def find_Area(radius):
    return math.pi * radius * radius

r = float(input(' Please Enter the radius of a circle: '))

diameter = find_Diameter(r)
circumference = find_Circumference(r)
area = find_Area(r)

print("\n Diameter Of a Circle = %.2f" %diameter)

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print(" Circumference Of a Circle = %.2f" %circumference)
print(" Area Of a Circle = %.2f" %area)
print ("")
```

```
print("Program 25")
def areaRectangle(a, b):
    return (a * b)
```

```
def perimeterRectangle(a, b):
    return (2 * (a + b))
```

```
a = 5
b = 6
print ("Area = ", areaRectangle(a, b))
```

```
print ("Perimeter = ", perimeterRectangle(a, b))
print ("")
```

```
print("Program 26")
import random
import math
# Taking Inputs
lower = int(input("Enter Lower bound:- "))
```

```
# Taking Inputs
upper = int(input("Enter Upper bound:- "))
```

```
# generating random number between
# the lower and upper
x = random.randint(lower, upper)
print("\n\tYou've only ",
      round(math.log(upper - lower + 1, 2)),
      " chances to guess the integer!\n")
```

```
# Initializing the number of guesses.
count = 0
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```
# for calculation of minimum number of
# guesses depends upon range
while count < math.log(upper - lower + 1, 2):
    count += 1
```

```
    # taking guessing number as input
    guess = int(input("Guess a number:- "))
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```
    # Condition testing
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    if x == guess:
        print("Congratulations you did it in ",
              count, " try")
        # Once guessed, loop will break
        break
    elif x > guess:
        print("You guessed too small!")
    elif x < guess:
        print("You Guessed too high!")

# If Guessing is more than required guesses,
# shows this output.
if count >= math.log(upper - lower + 1, 2):
    print("\nThe number is %d" % x)
    print("\tBetter Luck Next time!")
print ("")

print("Program 27")
import datetime

date=str(input('Enter the date(for example:09 02 2019):'))
day_name= ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday',
'Saturday','Sunday']
day = datetime.datetime.strptime(date, '%d %m %Y').weekday()
print(day_name[day])
print ("")

print("Program 28")
def find_missing(lst):
    return [x for x in range(lst[0], lst[-1]+1)
if x not in lst]

# Driver code
lst = [1, 2, 4, 6, 7, 9, 10]
print(find_missing(lst))
print ("")

print("Program 29")
char_list = ["a", "b" ,"c"]
string = "abcd"
matched_list = [characters in char_list for characters in string]
print(matched_list)
string_contains_chars = all(matched_list)
print(string_contains_chars)
print ("")

```

```
print("Program 30")
total = 0
evenSums = 0
oddSums = 0

evenCount = 0
oddCount = 0

done = False

while not done:
    user_in = input("Give me an integer or type 'done' to be done: ")

    if user_in.lower() == "done":
        done = True
    else:
        num = int(user_in)
        total += num

        if num % 2 == 0:
            evenSums += num
            evenCount += 1
        else:
            oddSums += num
            oddCount += 1

evenAverage = evenSums / evenCount if evenCount > 0 else 0
oddAverage = oddSums / oddCount if oddCount > 0 else 0

print("\nTotal sum:", total)
print("Even Average: " + str(evenAverage))
print("Odd Average: " + str(oddAverage))
```